

Lab Task 7: POS Tagging using Hidden Markov Model

Aim:

To implement Part-of-Speech (POS) Tagging using Hidden Markov Model (HMM) with the help of Python libraries such as Pandas, NLTK, spaCy, and Gensim.

Code:

```
from collections import defaultdict, Counter

# ♦ Sample tagged data (instead of treebank)

data = [

    [("The", "DT"), ("market", "NN"), ("is", "VBZ"), ("rising", "VBG")],

    [("She", "PRP"), ("is", "VBZ"), ("happy", "JJ")],

    [("Dogs", "NNS"), ("are", "VBP"), ("running", "VBG")]

]

# ♦ Transition & Emission Counts

trans = defaultdict(Counter)

emit = defaultdict(Counter)

for sent in data:

    prev = "<s>"

    for word, tag in sent:

        trans[prev][tag] += 1

        emit[tag][word] += 1

        prev = tag

# ♦ Tagging function

def tag_sentence(sentence):

    result = []

    for word in sentence:

        best_tag = max(emit, key=lambda t: emit[t][word])

        result.append((word, best_tag))

    return result

# Test
```

```
sentence = ["The", "market", "is", "rising"]
```

```
print("Tagged Sentence:")
```

```
print(tag_sentence(sentence))
```

Output:

Tagged Sentence:

```
[('The', 'DT'), ('market', 'NN'), ('is', 'VBZ'), ('rising', 'VBG')]
```